MIL-C-23212B(EC)
5 March 1971
SUPERSEDING
MIL-C-23212A(EC)
15 August 1969
(See 6.6)

MILITARY SPECIFICATION

COMMUNICATIONS SYSTEM, FLIGHT DECK, AN/SRC-22(V)

1. SCOPE

1.1 The equipment covered by this specification shall provide two-way radio communication between various fixed points and the flight deck of an aircraft carrier by means of frequency modulated radio telephony. When installed, the radio set shall be suitable for the following uses:

(a) Communication between the Flight deck control office (FDCO) and properly equipped deck personnel.
(b) Communication between Primary flight control (Pri-Fli) and properly equipped deck personnel.
(c) Retransmission of crash alarm and warning messages from a Type 5 MC paging system to FDCO, Pri-Fli, and properly equipped deck personnel.
(d) Automatic or manual retransmission of messages originated by equipped deck personnel to other properly equipped deck personnel.

1.2 Classification. - The equipment covered by this specification shall consist of the following items:

Items	Reference paragraph
1 - Radio set, fixed shipboard	3.4
2a - Radio set, helmet	3.5
2b - Radio set, handset	3.6

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-576 - Box, Wood, Veneer, Paper Overlaid
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Box, Fiberboard

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-D-3464 - Desiccants Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-Q-9858 - Quality Program Requirements
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water Vaporproof or Waterproof, Flexible
MIL-P-15024 - Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of.

FSC 5820

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STANDARDS

MILITARY

MIL-STD-105 - Sampling procedures and tables for inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets Skids, Runners, or Pallet-Type Base
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment
MIL-STD-200 - Electron Tubes; Selection and Use of.
MIL-STD-415 - Test Points and Test Facilities, for Electronic Systems and Associated Equipment; Design Standard for.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-701 - Lists of Standard Semiconductor Devices
MIL-STD-756 - Reliability Prediction
MIL-STD-781 - Reliability Tests; Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606).

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036).

3. REQUIREMENTS

3.1 First article inspection.- Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.2.1).

3.2 Parts and materials.- In the selection of parts and materials, fulfillment of the major design objectives shall be the prime consideration. In so doing, the following shall govern.

(a) Parts and materials, as approved by MIL-E-16400, shall be given first consideration.
(b) When the characteristics of standard parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts

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shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(c) When a non-standard part is used, the meeting of general material requirements such as non-flammable, fungus resistant, non-toxic, and so forth and the meeting of environmental conditions, as required by MIL-E-16400, shall be of prime consideration.

(d) In the selection of parts, the above considerations shall apply if the assemblies are to be built for detail part maintenance. In the event that an assembly or subassembly is built to be expendable, it shall be so constructed, and the quality of the individual parts shall be such, that the unit as a whole will pass material, electrical and environmental tests.

3.2.1 Nonstandard parts and material approval.- Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-16400, except as follows:

(a) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of MIL-E-16400 are not met.

(b) Electrical parts shall be submitted as nonstandard parts if:

(1) A standard part of the same size and weight exists; or,

(2) A standard part of the same size and weight does not exist and the performance characteristics of the nonstandard part are inferior to the nearest standard part for the intended application.

3.2.2 Tubes and semiconductor types.- First consideration shall be given to tubes and transistors listed in MIL-STD-200 and MIL-STD-701. The command or agency concerned shall be notified of all unlisted tubes and semiconductors. Approval is not required except under the conditions of 3.2.1(b).

3.2.3 Finishes.- Brass hardware and shafts may be used without a protective finish where the part is subject to wear.

3.2.4 Inductors.- Every precaution shall be taken to maintain the stability of "Q" of radio frequency inductors. The requirements shall be only those which enable meeting the equipment performance requirements.

3.2.5 Soldering (electrical connections).- Small lightweight parts or tubes employed in printed wiring assemblies which are processed by dip soldering techniques may use solder for mechanical strength provided adherence to the equipment performance requirements is not compromised.

3.2.6 Reliability.-

3.2.6.1 Quantitative reliability requirements.- The specified mean-time-between-failure (MTBF) θ_o, as defined in MIL-STD-781, of each item of this system shall be as follows:

Item	MTBF (θ_o) - Hours
Radio set, fixed shipboard	830
Radio set, helmet	1655
Radio set, handset	2070

3.2.6.2 Reliability program plans (see 6.1.2).- The contractor(s) shall submit, for approval by the command or agency concerned, a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785 for each item listed in 3.2.6.1. In the event that a single contractor is awarded a contract encompassing all of these items, he may submit only one reliability program plan. However, it must address each end item as a separate entity.

3.2.6.3 Reliability predictions (see 6.1.2).- The contractor(s) shall submit, for approval by the command or agency concerned, the following reliability predictions for each end item:

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(a) Feasibility prediction shall be made in accordance with the requirement of MIL-STD-756.
(b) Design prediction based on stress factors and part populations in accordance with the requirements of MIL-HDBK-217.

3.2.6.4 Source of failure rates.- The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned, when such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.2.6.5 Reliability reports.- Reports in accordance with MIL-STD-785 and also providing a complete accounting of progress on each element defined by the reliability program plan, results achieved and status of actions to resolve problems and correct deficiencies shall be submitted to the command or agency concerned (see 6.1.2).

3.2.6.6 Reliability demonstration tests.- The reliability testing shall be in accordance with 4.5.

3.2.7 Antenna frequencies and helmet sizes.- The contractor shall request antenna frequencies for those equipments requiring crystals and helmet sizes for the headset.

3.3 Design and construction.-

3.3.1 General requirements.- The system shall meet the applicable requirements of MIL-E-16400, except as hereinafter specified.

3.3.2 Blown fuse indicators.- Blown fuse indicators will not be required.

3.3.3 Test points.- The number and location of test points shall be adequate for rapid trouble shooting with standard Navy test equipment. A list of such test points shall be submitted to the command or agency concerned for approval. MIL-STD-415 shall be used as a guide, where applicable. (see 6.1.2)

3.4 Radio set, fixed shipboard.-

3.4.1 The equipment in this radio set shall consist of the following:

Item	Max. allow size (Cu. in.)	Max. allow wgt (lbs.)	Applicable paragraph
(a) Handset	NA	2	3.4.2
(b) Speaker	1500	12	3.4.3
(c) Control unit	1000	15	3.4.4
(d) Base station unit	16,000	200	3.4.5
(e) Antenna	NA	15	3.4.6
(f) Battery charger	2000	35	3.4.7

3.4.1.1 Standard test conditions.- The following conditions shall be used to establish normal performance characteristics under standard test conditions and for making laboratory bench tests.

(a) Temperature +20°C. to 35°C.
(b) Altitude Normal ground level.
(c) Vibration None.
(d) Humidity Prevailing room ambient.
(e) Input power voltage 117 ±3v 1ϕ 60 hertz (Hz).
(f) Signal Source - Calibrated signal generator whose internal plus external resistance is 50 ohms, connected to the receiver in series with a 50 ohm 6 dB pad and the voltage input to the receiver measured at the input terminals of the 6 dB pad. (hard microvolts). For tests, except where otherwise specified, the frequency of the generator shall be adjusted to center frequency ±1 KHz on which the receiver is intended to operate.

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(g) Modulation 5 kilohertz (kHz) deviation at 1000 Hz.
(h) Output load receiver 600 ohms resistive.
transmitter 50 ohms resistive.
(i) Squelch fully unsquelched.

3.4.1.2 Service conditions.- The equipment shall operate satisfactorily under any of the service conditions for class 4 equipments or combinations of service conditions as specified in MIL-E-16400.

3.4.1.3 Shock.- The equipment shall be capable of withstanding the shock test for Grade A, Type B, Class I, lightweight equipment as specified in MIL-S-901. Shock mounts shall not be used.

3.4.1.4 Vibration.- Shipboard equipment meeting the requirements of MIL-STD-167 shall be identified by securely attaching an identification plate to each equipment in accordance with one of the following:

This equipment tested under MIL-STD-167, Type I without mounts to 50 Hz

or

Duplicate equipment tested under MIL-STD-167, Type I without mounts to 50 Hz.

The identification plates shall be in accordance with Type A, B, or C of MIL-P-15024.

3.4.1.5 Inclination.- The equipment shall meet the inclination requirements of MIL-E-16400.

3.4.1.6 Enclosures.- Enclosures shall be "drip-proof" in accordance with MIL-STD-108.

3.4.1.7 Elapsed time meters.- Elapsed time meters shall not be required.

3.4.2 Transducer handset.- The handset shall have the following characteristics:

(a) A push-to-talk switch shall be located in the handle of the handset.
(b) A coil cord shall be provided.

3.4.2.1 Microphone.- A dynamic noise canceling microphone shall be employed and shall have a frequency response of from 300 to 6000 Hz at a nominal impedance of 15 ohms.

3.4.2.2 Earphone.- A dynamic earphone element shall be employed and shall have a frequency response of from 300 to 6000 Hz at a nominal impedance of 5 ohms.

3.4.3 Loudspeaker.- A cone-type weatherproof loudspeaker shall be employed. It shall have the following electrical characteristics:

(a) Power capability	4 watts.
(b) Frequency response	At least 300 to 5000 Hz.
(c) Dispersion	360°.
(d) Impedance	8 ohms.

3.4.4 Control unit.- The control unit shall be enclosed in a "drip-proof" case in accordance with MIL-STD-108. All control units shall be identical, to permit interchangeability. Included in the control unit are the following functional units: microphone amplifier, speaker amplifier, headphone amplifier, power supply.

3.4.4.1 Microphone amplifier.- The microphone amplifier shall have a frequency response within ±3 dB from 300 to 6000 Hz. The input impedance shall be nominal 15 ohms. The output shall produce an 0-dBm signal across a balanced 600 ohm line.

3.4.4.2 Speaker amplifier.- The speaker amplifier shall be capable of delivering 8 watts of audio at 1000 Hz to an 8-ohm resistive load. The amplifier shall be capable of delivering the rated output with 0 dBm input to the control unit from a balanced 600 ohm line. The frequency response shall be 300 to 5000 Hz ± 3 dB, with a maximum 1000 Hz distortion of 5 percent at rated output.

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3.4.4.3 Handset amplifier.- The handset amplifier shall be capable of delivering 20 milliwatts (mW) of audio at 1000 Hz into its rated load impedance. The frequency response shall be 300 to 6000 Hz ± 3 dB with a maximum 1000 Hz distortion of 5 percent at 20 mW.

3.4.4.4 Power supply.- The power supply shall operate from a primary power source of 105 to 125 volts, 60 Hz alternating current (ac) and shall provide all the voltages necessary to power the control unit amplifiers.

3.4.4.5 Switching functions.- The control unit shall provide the following switching functions.

3.4.4.5.1 Loudspeaker muting.- The loudspeaker shall be muted when the local microphone push-to-talk switch is operated.

3.4.4.5.2 Power switch.- The power switch shall operate the control unit power supply.

3.4.4.5.3 Transmitter keying.- The microphone push-to-talk switch shall key the transmitter in accordance with priority status. (See 3.4.5.3).

3.4.4.5.4 Retransmission.- The operator of the flight deck control unit shall be able to select the desired mode of retransmission manual or automatic.

3.4.4.6 Front panel controls, indicators, and connectors.- Front panel controls, indicators, and connectors shall include:

(a) Volume control for monitor speaker.
(b) Volume control for monitor handset.
(c) Busy light, dimmer type - shall indicate when circuit is in use by another station.
(d) Transmit light, dimmer type - shall indicate whenever control unit is transmitting.
(e) Power-on light, dimmer type.
(f) Retransmission selection switch.
(g) Power-on switch.

3.4.4.7 Additional controls and connections.- Additional controls and connections shall be as specified in 3.4.4.7.1 and 3.4.4.7.2.

3.4.4.7.1 Side panel location.-

(a) Interconnecting cable connector.
(b) AC line input connector.
(c) Handset connector.
(d) Speaker connector.

3.4.4.7.2 Chassis location.- Screwdriver operated controls to adjust:

(a) Minimum volume for handset.
(b) Minimum volume for speaker.
(c) Microphone amplifier gain.
(d) Input level adjustment.

3.4.5 Base station unit.- The base station unit shall include the following functional units: compressor amplifier, confidence tone generator and five-second delay, automatic selector, receiver, transmitter, and power supply. Control cables from the two control units shall be terminated in this unit.

3.4.5.1 Compressor-amplifier.- The compressor-amplifier shall have a frequency response of 300 to 6000 Hz ± 3 dB and shall provide 0 dBm audio output levels into a balanced line. Compression characteristics shall be such that a 30 dB change in input level will cause not more than a 6 dB change in output level. The compressor output drives the two control units and the transmitter modulator. The amplifier input shall be selected through the automatic selector from one of the following:

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(a) Primary flight microphone amplifier output.
(b) Flight deck control microphone amplifier output.
(c) Fixed station receiver audio output.
(d) Crash warning audio.

3.4.5.2 Confidence tone generator and five-second delay.- The confidence tone generator shall consist of an audio oscillator and a timing device which shall apply a 250-500 millisecond, 800-1000 Hz tone to the audio line approximately every 10 seconds. It shall also key the transmitter so the confidence tone is received by the portable units. The confidence tone shall not be initiated unless the transmitter is radiating radio frequency (rf) energy and shall be disabled during any voice or crash warning transmission. The five-second delay is designed to give the crash alarm first priority for the first five seconds after it is switched on; after five seconds the crash alarm shall take last priority.

3.4.5.3 Automatic selector.- The automatic selector shall provide distribution of signals according to the following priority status:

(a) Crash alarm and audio signals from the type 5MC system shall have top priority for 5 seconds after a crash alarm is initiated.
(b) Primary flight control.
(c) Flight deck control officer.
(d) Mobile units.
(e) Signals from the type 5 MC system, after the first 5 seconds. (Unless alarm is reinitiated, at which time signals shall again have top priority for 5 seconds).

3.4.5.4 Receiver.- The receiver shall be a transistorized unit mounted in a shielded enclosure to minimize radio frequency interference.

3.4.5.4.1 RF input impedance.- The receiver input circuits shall be designed to match a 50 ohm coaxial transmission line with a VSWR not to exceed 3:1.

3.4.5.4.2 Sensitivity.- The receiver shall be capable of 12 dB signal plus noise-to-noise ratio for an input not exceeding two microvolts. The audio power output at sensitivity shall be not more than 3 dB down from power output at 1000 microvolts, and ±5 kHz deviation.

3.4.5.4.3 Frequency.- The receiver shall be capable of reception of any one frequency channel within the range of 132.0 to 150.8 MHz.

3.4.5.4.4 Oscillator stability.- Crystal oscillator stability of ± .003 percent shall be maintained.

3.4.5.4.5 Modulation acceptance bandwidth.- With an rf input signal of 4 microvolts, the receiver shall accept a signal with a deviation of at least ±7.5 kHz.

3.4.5.4.6 Selectivity.- The response to signals differing in frequency by less than 17.5 kHz from the center frequency shall not be more than 6 dB down from the response to signals at the center frequency. The response to signals differing in frequency by 75 kHz or more from the center frequency shall be at least 60 dB down from the response to signals at center frequency.

3.4.5.4.7 Image rejection.- The response of the receiver to any image frequency shall be not less than 50 dB below the response to signals at the center frequency.

3.4.5.4.8 Squelch.- The receiver shall have a suitable squelch control which shall permit adjustment of the threshold level over a practical squelching range. The squelch control shall quiet the receiver when there is no rf signal present. Signal inputs of 2 microvolts shall automatically unsquelch a receiver set at the threshold point.

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3.4.5.4.9 Audio power output.- The receiver output shall be at least 1 mW into a 600-ohm load with 1000 microvolts (uV) carrier input signal deviated ±7.5 kHz at 1000 Hz. The output shall rise not more than 3 dB when the input rises to 100,000 uV.

3.4.5.4.10 Audio frequency response.- The overall receiver audio frequency response shall not vary more than +1 to -3 dB over the frequency range of 300 to 6000 Hz using 1000 Hz as reference.

3.4.5.4.11 Hum and noise ratio.- With a 1000 uV unmodulated signal applied to the receiver, the hum and noise level shall be at least 40 dB below the 1 mW power output level of the receiver.

3.4.5.4.12 Audio distortion.- The overall receiver audio distortion at 1000 Hz for a power output of 1 mW shall not exceed 5 percent.

3.4.5.4.13 Undesired conducted power.- Radio interference voltage appearing on the external power conductors shall not exceed the limits of class 1A and 1B of MIL-STD-461.

3.4.5.4.14 Undesired radiated power from the receiver (transmitter on standby) shall be 30 microwatts, or less, at the antenna.

3.4.5.4.15 Environmental limits.- At the service condition extremes specified in MIL-E-16400, degradation from specified performance of the receiver will be permitted as follows:

(a)	Sensitivity	- limit 10 uV
(b)	Squelch	- limit 10 uV
(c)	Modulation acceptance BW	- no more than 20 percent decrease
(d)	Audio power output	- 3 dB
(e)	Audio distortion	- 10 percent at degraded power output

3.4.5.5 Transmitter.- The transmitter shall be a transistorized unit mounted in a shielded enclosure to minimize radio frequency interference.

3.4.5.5.1 RF power output.- The transmitter power output shall be not less than 300 mW when measured across a 50 ohm resistive load.

3.4.5.5.2 Frequency.- The transmitter frequency shall be crystal controlled. The transmitter shall be capable of operation on any frequency between 132 to 150.8 MHz.

3.4.5.5.3 Frequency accuracy.- An overall frequency accuracy of ± 0.003 percent shall be maintained.

3.4.5.5.4 Carrier wave purity.- The fundamental output shall be free from seriously interfering harmonics, switching noise, and other objectionable radiated energy. Harmonics of the carrier shall be attenuated at least 30 dB below the carrier. Spurious radiations shall be attenuated at least 30 dB below the carrier.

3.4.5.5.5 Modulation.- Phase or frequency modulation shall be employed. The system shall be designed for a maximum frequency deviation of ±7.5 kHz at 1000 Hz modulation.

3.4.5.5.6 Audio frequency response.- The transmitter audio frequency response, with limiters disabled, shall not vary more than +1 to -3dB between 300 and 6000 Hz using 1000 Hz as reference.

3.4.5.5.7 Audio frequency distortion.- The overall transmitter audio distortion with a 1000 Hz modulating signal shall not exceed 10 percent at ± 5 kHz deviation.

3.4.5.5.8 Audio input impedance.- The nominal modulator input impedance shall be 600 ohms.

3.4.5.5.9 Audio input voltage.- The modulator gain shall be adjustable so that a deviation of ± 7.5 kHz may be achieved at an input level of 0 dBm at 1000 Hz.

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3.4.5.5.10 Environmental limits. Environmental limits shall be as follows:

(a) Power output	- 3 dB
(b) Modulation deviation	- 3 dB
(c) Audio distortion	- 15 percent.

3.4.5.6 Power supply.- The power supply shall operate from a primary power source of 105 to 125 volts. 60 Hz ac and shall provide the necessary voltages to operate all the auxiliary control circuitry.

3.4.5.6.1 Overload protection.- The power supply shall be fused to protect against overload.

3.4.5.7 Front panel controls and indicators.- The following controls and indicators shall be located on the front panel:

(a) Master "power-on" switch.
(b) "Power-on" light.
(c) Fuse.

3.4.6 Antenna.- The antenna shall exhibit the following characteristics:

(a) Radiation pattern	- omnidirectional.
(b) Impedance	- 50 ohms, nominal.
(c) Frequency range	- 130 to 150.8 MHz with a VSWR not greater than 1.5:1

3.4.7 Battery charger.- The battery charger shall operate from the ship's 115 Vac supply with an on-off switch, line fuse and an "on" indicating lamp. The charger shall furnish essentially constant charging current to charge from 1 to 20 batteries simultaneously and protection shall be provided to avoid over-charge damage to the batteries. The charging time from a fully discharged to a fully charged condition shall be 24 hours. A test facility shall be provided on the charger unit that will indicate go no-go readings for service condition of batteries.

3.5 Radio set helmet.-

3.5.1 General requirements.- The equipment in the radio set shall consist of the following:

Item	Max allow size (inches)	Max allow weight (lbs.)	Reference paragraph
(a) Headset	-	4.3 lb.	3.5.2
(b) Antenna	20	0.25 lb.	3.5.3
(c) Battery	1.4 diam x 2.5 long	0.25 lb.	3.5.4
(d) Frame, helmet	-	1.40 lb.	3.5.5
(e) Transmitter	4.9 x 3.5 x 2.3	1.50 lb.	3.5.6
(f) Receiver	4.9 x 3.5 x 2.3	1.50 lb.	3.5.7

3.5.1.1 Standard conditions.- The following conditions shall be used to establish normal performance characteristics under standard conditions and when making laboratory bench tests.

(a) Temperature	+20°C to 35°C.
(b) Altitude	Normal ground level.
(c) Vibration	None.
(d) Humidity	Prevailing room ambient.
(e) Input voltage	Nominal battery voltage 8.4 volts.
(f) Same as 3.4.1.1	(f).
(g) Same as 3.4.1.1	(g).
(h) Output load	Receiver 10 ohms resistive. Transmitter 50 ohms resistive.
(i) Same as 3.4.1.1	(i).

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3.5.1.2 Service conditions.- The equipment shall operate satisfactorily under any of the service conditions for class 2 equipment, or combinations of service conditions as required in MIL-E-16400.

3.5.1.3 Shock.- The equipment shall be capable of withstanding the shock test for Grade A, Type B, Class I, lightweight equipment as specified in MIL-S-901. Shock mounts shall not be used.

3.5.1.4 Vibration.- Shipboard equipment meeting the requirements of MIL-STD-167 shall be identified by securely attaching an identification plate to each equipment in accordance with one of the following:

This equipment tested under MIL-STD-167, Type I without mounts to 50 Hz

or

Duplicate equipment tested under MIL-STD-167, Type I without mounts to 50 Hz.

The identification plates shall be in accordance with Type A, B, or C of MIL-P-15024.

3.5.1.5 Inclination.- The equipment shall meet the inclination requirements of MIL-E-16400.

3.5.1.6 Enclosures.- Enclosures shall be "drip-proof" from any attitude in accordance with MIL-STD-108.

3.5.2 Headset.- The headset shall consist of the receiver-transmitter components, battery, and earphones mounted in ear protector pods having an attached microphone. The earpods shall be joined by a headband which will also support the antenna. The headset shall be attached to, but capable of functioning independent of, the protective helmet. A press-to-talk switch shall be located on the left earpod.

3.5.2.1 Headband.- The headband shall be of the spring-loaded type and adjustable to provide fitting for the variation in head sizes. A standby provision for quiet area use shall be incorporated to permit the release of sealing force around the ears without the removal of the headset or loss of communications.

3.5.2.2 Noise reduction.- The ear protectors shall attenuate ambient noise 20 dB or more between 300 and 6000 Hz without the use of earplugs. Sealing force applied to the ear protectors shall be sufficient to provide the required attenuation at the sealing face. The pinna of the ear shall not be deformed by the ear protector.

3.5.2.3 Material and design.- The combined ear protector-radio enclosure shall be of high impact resisting plastic or equal. It shall be possible to remove the outer covering for test and maintenance without the disconnection or disassembly of electrical components from the headset.

3.5.2.4 Center of gravity.- The headset weight approximating 4.3 pounds shall be balanced and located near or below the center of gravity of the head. Projection from the contour of the head shall be minimized to reduce the sail-effect in wind of 35 to 40 knots.

3.5.3 Antenna.- The antenna shall be mounted on the headband in such a manner as to provide minimum interference or discomfort to the wearer. It shall be designed for broadband operation over the frequency range 132.0 to 150.8 MHz.

3.5.4 Battery.- Sealed, rechargeable, nickel-cadmium batteries shall be employed.

3.5.4.1 Life.- Battery voltage shall not drop more than 20 percent from nominal voltage in 10 hours under the following duty cycle:

(a) Transmit (5 percent).
(b) Receive (100 mW audio) (5 percent).
(c) Receive (1 mW audio) (12 percent).
(d) Receive (squelch) (78 percent).

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3.5.5 Helmet.-

3.5.5.1 Impact protection.- The helmet shall provide bump protection. It shall consist of a semirigid plastic shell lined with a shock absorbent nonresilient material. The form fitting characteristics of the helmet shall distribute the weight of the headset to avoid creating undue sensitive pressure points. Position stability shall be maintained by means of an adjustable chin strap.

3.5.5.2 Size adjustment.- The necessary inventory of head sizes shall be held to a minimum. Provisions shall be made for individual fit adjustment within selected head size groups.

3.5.5.3 Eye protection.- The headset and associated protective helmet shall permit the use of standard Type II flying goggles.

3.5.6 Transmitter.- The transmitter shall be a transistorized unit mounted in an ear pod. The transmitter shall exhibit the following characteristics under standard conditions.

3.5.6.1 Power output.- The transmitter rf power output shall be not less than 5 mW when measured across a 50-ohm resistive load.

3.5.6.2 Frequency.- The transmitter frequency shall be crystal controlled. The transmitter shall be capable of operation on any frequency within the range of 132.0 to 150.8 MHz.

3.5.6.3 Frequency accuracy.- An over-all frequency accuracy of ±0.003 percent shall be maintained.

3.5.6.4 Carrier wave purity.- The fundamental output shall be free from seriously interfering harmonics, switching noise, and other objectionable radiated energy. Harmonics of the carrier shall be attenuated at least 20 dB below the carrier level.

3.5.6.5 Modulation.- Phase or frequency modulation shall be employed. The system shall be designed for a maximum frequency deviation of ± 5 kHz at 1000 Hz modulation.

3.5.6.6 Audio frequency response.- The transmitter audio frequency response, with limiters disabled, shall not vary more than +1 to -3 dB between 300 and 6000 Hz using 1000 Hz as reference.

3.5.6.7 Audio frequency distortion.- The overall transmitter audio distortion with a 1000 Hz modulating signal shall not exceed 10 percent at ± 5 kHz deviation.

3.5.6.8 Audio input impedance.- The modulator input impedance shall be designed to match the microphone used.

3.5.6.9 Audio input voltage.- The modulator gain shall be adjustable so that a deviation of ± 7.5 kHz may be achieved with rated microphone output voltage at 1000 Hz.

3.5.6.10 Clipper.- A clipper shall be employed in the speech circuits.

3.5.6.11 Microphone.- A dynamic noise-canceling microphone shall be employed and shall have a frequency response of 300 to 5000 Hz.

3.5.6.12 Environmental limits.- At the service condition extremes for class 2 equipment, of MIL-E-16400 degradation from specified performance of the transmitter shall be permitted as follows:

(a) Power output - 3 dB.
(b) Modulation deviation - 3 dB.
(c) Audio distortion - 15 percent.

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3.5.7 Receiver.- The receiver shall be a transistorized unit mounted in an ear pod. It shall exhibit the following characteristics under standard conditions.

3.5.7.1 Rf input impedance.- The receiver input circuits shall be designed for connection to the antenna specified in 3.5.3.

3.5.7.2 Sensitivity.- The receiver shall be capable of 12 dB signal plus noise-to-noise ratio for an input not exceeding 5 microvolts into 50 ohms. The audio power output at sensitivity will not be more than 3 dB down from power output at 1000 microvolts and ±5 kHz deviation.

3.5.7.3 Frequency.- The receiver shall be capable of reception of any one frequency channel within the range of 132.0 to 150.9 MHz.

3.5.7.4 Oscillator stability.- Crystal oscillator stability of ±0.003 percent shall be maintained.

3.5.7.5 Modulation acceptance bandwidth.- With an rf input signal of 10 microvolts, the receiver shall accept a signal with a deviation of at least ±7.5 kHz.

3.5.7.6 Selectivity.- The response to signals differing in frequency by less than 17.5 kHz from the center frequency shall be not more than 6 dB down from the response to signals at the center frequency. The response to signals differing in frequency by 75 kHz or more from the center frequency shall be at least 50 dB down from the response to signals at center frequency.

3.5.7.7 Image rejection.- The response of the receiver to any image frequency shall be not less than 50 dB below the response to signals at the center frequency.

3.5.7.8 Squelch.- The receiver shall have a suitable squelch control which will permit adjustment of the threshold level over a practical squelching range. The squelch control shall quiet the receiver when there is no rf signal present. Signal inputs of 5 microvolts shall automatically unsquelch a receiver set at the threshold point.

3.5.7.9 Audio power output.- The audio output circuits shall deliver at least 2 mW of power to the earphones with a 1000 microvolt carrier input signal deviated ±7.5 kHz at 1000 Hz. Output shall not rise more than 3 dB when the input rises to 100,000 microvolts.

3.5.7.10 Audio frequency response.- The overall receiver audio frequency response shall not vary more than +1 to -3 dB over the frequency range of 300 to 6000 Hz using 1000 Hz as reference.

3.5.7.11 Audio distortion.- The overall receiver audio distortion at 1000 Hz for a power of 100 mW shall not exceed 10 percent.

3.5.7.12 Noise actuated automatic volume control.-

3.5.7.12.1 Sensing microphone. A sensing microphone, located in the receiver and exposed to ambient noise, shall be used to provide an indication in the receiver proportional to the sound level of the ambient noise.

3.5.7.12.2 Automatic volume control (AVC) characteristics.- With the manual VC on the earpod adjusted to maximum output, and when the ambient noise is 110 dB or less the audio power available to the earphones shall be 2 mW ±1 dB. When the ambient noise is 130 dB or more the audio power available to the earphones shall be 100 mW -1 + 2 dB. The audio power available to the earphones under the conditions described above shall be essentially linear from the end point values specified herein. These measurements to be made with an input signal of 1000 uV with a deviation of ±7.5 kHz at 1000 Hz.

3.5.7.13 Undesired radiated power.- Undesired radiated power from the receiver (transmitter on standby) shall be 30 microwatts or less at the antenna.

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3.5.7.14 Earphones.- The earphones shall have a frequency response of 300 to 6000 Hz at a nominal impedance of 10 ohms.

3.5.7.15 Environmental limits.- At the service condition extremes specified for class 2 equipment in MIL-E-16400 degradation from specified performance of the receiver will be permitted as follows:

(a) Sensitivity - limit 10 microvolts.
(b) Squelch - limit 10 microvolts.
(c) Modulation acceptance BW - no more than 20 percent decrease.
(d) Audio power output - 3 dB degradation.
(e) Audio distortion - 15 percent at degraded maximum power.

3.5.7.16 Controls.- The operator controls on the helmet radio set shall be an on-off switch, volume control and transmit-receive switch.

3.6 Radio set, handset.-

3.6.1 Parts.- Parts in this radio set shall consist of the following:

Item	Paragraph reference
(a) Microphone	3.6.2
(b) Earphone	3.6.3
(c) Battery	3.6.4
(d) Frame	3.6.5
(e) Antenna	3.6.6
(f) Receiver-transmitter	3.6.7

3.6.1.1 Standard conditions.- Standard conditions shall be as specified in 3.5.1.1.

3.6.1.2 Service conditions.- Service conditions shall be as specified in 3.5.1.2.

3.6.1.3 Shock, vibration and inclination.- Shock, vibration and inclination shall be as specified in 3.5.1.3, 3.5.1.4, and 3.5.1.5 respectively.

3.6.1.4 Enclosures.- Enclosures shall be as specified in 3.5.1.6.

3.6.2 Microphone.- The microphone shall be a noise-canceling type, having a frequency response of 300 to 5000 Hz.

3.6.3 Earphone.- The earphone shall have a frequency response of 300 to 6000 Hz at a nominal impedance of 5 ohms.

3.6.4 Battery.- The battery shall be as specified in 3.5.4.

3.6.5 Frame, handset.-

3.6.5.1 Material and design.- High impact resisting plastic, or equal, shall be used. An on-off switch, volume control, and a transmit-receive switch shall be conveniently located for operator control.

3.6.6 Antenna.- A telescoping antenna shall be used.

3.6.7 Receiver and transmitter.- The receiver and transmitter shall be completely transistorized.

3.6.7.1 Transmitter.- The transmitter shall be as specified in 3.5.6.

3.6.7.2 Receiver.- The receiver shall be as specified in 3.5.7, except for 3.5.7.1 and 3.5.7.12, which shall be deleted.

3.6.7.2.1 The receiver input circuits shall be designed for connection to the antenna specified in 3.6.6.

3.7 Synchros.- Synchros are not required.

3.8 Hydraulic shock.- Hydraulic shock requirements are not applicable.

3.9 Workmanship.- Workmanship shall be in accordance with MIL-E-16400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement.- The contractor shall provide and maintain a written quality program acceptable to the command or agency concerned for supplies covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.1.1 Quality plan.- A quality Plan, applicable to the equipments covered by this specification, shall be prepared for approval by the command or agency concerned.

4.1.2 Government verification.- Quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification (not excluding any requirement for which detailed tests are not specified herein). Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures.- The contractor shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection to the command or agency concerned, via the Government inspector for review and approval prior to beginning tests. Each test shall be identified in accordance with 4.2 as to classification. (see 6.1.2)

4.1.3.1 Procedures for inspections, tests, demonstrations, and so forth shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.1.4 Quality assurance terms and definitions.- Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection.- The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental tests
(c) Reliability testing
(d) Inspection of preparation for delivery

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4.3 First article inspection.- Unless otherwise specified (see 6.1.2), one of each unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I - First Article Inspection

Requirement	Reference paragraph
Radio set, fixed shipboard	
1. Production inspection	4.4.1
2. Production control inspection	4.4.2
3. Environmental tests	4.4.3
4. Weights and dimensions	3.4.1
5. Transducer handset	3.4.2
6. Base station unit	
(a) Receiver	
(1) Rf input impedance	3.4.5.4.1
7. Controls and control circuits	See MIL-E-16400
8. Antenna	3.4.6
9. Salt spray	See MIL-E-16400
Radio set helmet	
1. Production inspection	4.4.1
2. Production control inspection	4.4.2
3. Environmental testing	4.4.3
4. Weights and dimensions	3.5.1
5. Headset	3.5.2
(a) Material and design	3.5.2.3
(b) Center of gravity	3.5.2.4
6. Antenna	3.5.3
7. Battery	3.5.4
8. Helmet	
(a) Impact protection	3.5.5.1
(b) Size adjustment	3.5.5.2
(c) Eye protection	3.5.5.3
9. Transmitter	
(a) Audio input impedance	3.5.6.8
10. Receiver	
(a) RF input impedance	3.5.7.1
(b) Noise actuated automatic volume control	
(1) Sensing microphone	3.5.7.12.1
(c) Controls	3.5.7.16
11. Salt spray	See MIL-E-16400
12. Controls and control circuits	See MIL-E-16400

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Table I - First Article Inspection (Continued)

Requirement	Reference paragraph
Radio set handset	
1. Production inspection	4.4.1
2. Production control inspection	4.4.2
3. Environmental tests	4.4.3
4. Frame, handset	
(a) Material and design	3.6.5.1
5. Antenna	3.6.6
6. Salt spray	See MIL-E-16400

4.3.1 First article inspection report.- A first article inspection report shall be submitted to the command or agency concerned covering the results of the contractor's first article inspection. (see 6.1.2)

4.4 Quality conformance inspection.

4.4.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from designs, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the tests specified in table II.

TABLE II - Production Inspection

Requirement	Reference paragraph
Radio set, fixed shipboard	
1. Transducer handset	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.2
2. Loudspeaker	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.2
3. Control unit	
(a) Surface examination	4.4.1.1
(b) Microphone amplifier	3.4.4.1
(c) Speaker amplifier	3.4.4.2
(d) Handset amplifier	3.4.4.3
(e) Switching functions	3.4.4.5
(f) Front panel controls, indicators and connectors	3.4.4.6
(g) Operating test	4.4.1.2
4. Base station unit	
(a) Surface examination	4.4.1.1
(b) Compressor-amplifier	3.4.5.1

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Table II - Production Inspection (Continued)

Requirement	Reference paragraph
(c) Confidence tone generator and five-second delay	3.4.5.2
(d) Automatic selector	3.4.5.3
(e) Receiver	
(1) Sensitivity	3.4.5.4.2
(2) Oscillator stability	3.4.5.4.4
(3) Modulation acceptance bandwidth	3.4.5.4.5
(4) Selectivity	3.4.5.4.6
(5) Image rejection	3.4.5.4.7
(6) Squelch	3.4.5.4.8
(7) Audio power output	3.4.5.4.9
(8) Audio frequency response	3.4.5.4.10
(9) Hum and noise ratio	3.4.5.4.11
(10) Audio distortion	3.4.5.4.12
(f) Transmitter	
(1) RF power output	3.4.5.5.1
(2) Frequency accuracy	3.4.5.5.3
(3) Carrier wave purity	3.4.5.5.4 and 4.4.1.3
(4) Modulation	3.4.5.5.5
(5) Audio frequency response	3.4.5.5.6
(6) Audio frequency distortion	3.4.5.5.7
(7) Audio input voltage	3.4.5.5.9
(g) Operating test	4.4.1.2
5. Battery charger	3.4.7
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.2
6. Antenna	
(a) Surface examination	4.4.1.1
Radio set, helmet	
1. Headset	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.5
2. Antenna	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.5
3. Battery	
(a) Surface examination	4.4.1.1
4. Helmet	
(a) Surface examination	4.4.1.1

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Table II - Production Inspection (Continued)

Requirement	Reference paragraph
5. Transmitter	
(a) Surface examination	4.4.1.1
(b) Power output	3.5.6.1
(c) Frequency accuracy	3.5.6.3
(d) Carrier wave purity	3.5.6.4 and 4.4.1.4
(e) Modulation	3.5.6.5
(f) Audio frequency response	3.5.6.6
(g) Audio frequency distortion	3.5.6.7
(h) Audio input voltage	3.5.6.9
(i) Operating test	4.4.1.5
6. Receiver	
(a) Surface examination	4.4.1.1
(b) Sensitivity	3.4.5.4.2
(c) Oscillator stability	3.5.7.4
(d) Modulation acceptance bandwidth	3.5.7.5
(e) Selectivity	3.5.7.6
(f) Image rejection	3.5.7.7
(g) Squelch	3.5.7.8
(h) Audio power	3.5.7.9
(i) Audio frequency response	3.5.7.10
(j) Audio distortion	3.5.7.11
(k) Automatic volume control (AVC) characteristics	3.5.7.12.1
(l) Operating test	4.4.1.5
Radio Set, handset	
1. Microphone	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.6
2. Earphone	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.6
3. Battery	
(a) Surface examination	4.4.1.1
4. Frame handset	
(a) Surface examination	4.4.1.1
5. Antenna	
(a) Surface examination	4.4.1.1
(b) Operating test	4.4.1.6
6. Transmitter	
(a) Surface examination	4.4.1.1
(b) Power output	3.5.6.1
(c) Frequency accuracy	3.5.6.3

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Table II - Production Inspection (Continued)

Requirement	Reference paragraph
(d) Carrier wave purity	3.5.6.4 and 4.4.1.6
(e) Modulation	3.5.6.5
(f) Audio frequency response	3.5.6.6
(g) Audio frequency distortion	3.5.6.7
(h) Audio input voltage	3.5.6.9
(i) Operating test	4.4.1.6
7. Receiver	
(a) Surface examination	4.4.1.1
(b) Sensitivity	3.5.7.2
(c) Oscillator stability	3.5.7.4
(d) Modulation acceptance bandwidth	3.5.7.5
(e) Selectivity	3.5.7.6
(f) Image rejection	3.5.7.7
(g) Squelch	3.5.7.8
(h) Audio power	3.5.7.9
(i) Audio frequency response	3.5.7.10
(j) Audio distortion	3.5.7.11
(k) Operating test	4.4.1.5

4.4.1.1 Surface examination.- Equipment shall be examined for workmanship, assembly and fit, and marking.

4.4.1.2 Operating test.- A functional operating test shall be performed to insure qualitatively the proper functioning of all components of radio set, fixed shipboard, including operating controls. A dummy load may be substituted for the antenna.

4.4.1.3 Carrier wave purity.- Tests shall be performed to verify that harmonics are at least 30 dB below the carrier. (This test is applicable to the base station).

4.4.1.4 Carrier wave purity.- Tests shall be performed to verify that harmonics are at least 20 dB below the carrier. (This test is applicable to the handset and headset transceivers).

4.4.1.5 Operating test.- A functional operating test shall be performed to insure qualitatively the proper functioning of all components of radio set, helmet including operating controls.

4.4.1.6 Operating test.- A functional operating test shall be performed to insure qualitatively the proper functioning of all components of radio set, handset including operating controls.

4.4.1.7 Standard test conditions.- Unless otherwise specified, standard test conditions as specified in 3.4.1.1; 3.5.1.1; and 3.6.1.1 shall be used in performing tests for each appropriate unit.

4.4.2 Production control inspection.- Production control inspection, including sampling, shall conform to table III below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of Production control inspection prior to release for shipment.

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TABLE III - Production Control Inspection

Inspection	Inspection paragraph	AQL's
Radio Set, fixed shipboard		
1. Transducer handset		
(a) Microphone	3.4.2.1	6.5
(b) Earphone	3.4.2.2	6.5
2. Loudspeaker	3.4.3	6.5
3. Control Unit	3.4.4	6.5
(a) Supply line voltage	3.4.4.4 and 4.4.2.1	6.5
4. Base station unit		
(a) Receiver		
(1) Frequency	3.4.5.4.3	6.5
(2) Undesired conducted power	3.4.5.4.13	*See Note 1
(3) Undesired radiated power	3.4.5.4.14	*See Note 1
(b) Transmitter		
(1) Frequency	3.4.5.5.2	6.5
(2) Carrier wave purity	3.4.5.5.4 and 4.4.2.3	*See Note 1
(c) Supply line voltage	3.4.5.6 and 4.4.2.2	6.5
Radio Set, Helmet		
1. Battery		
(a) Life	3.5.4.1	6.5
2. Transmitter		
(a) Frequency	3.5.6.2	6.5
(b) Carrier wave purity	3.5.6.4 and 4.4.2.4	*See Note 1
(c) Microphone	3.5.6.11	6.5
3. Receiver		
(a) Frequency	3.5.7.3	6.5
(b) Undesired radiated power	3.5.7.13	*See Note 1
(c) Earphones	3.5.7.14	6.5
Radio set, handset		
1. Microphone	3.6.2	6.5
2. Earphone	3.6.3	6.5
3. Battery		
(a) Life	3.5.4.1	6.5
4. Transmitter		
(a) Frequency	3.5.6.2	6.5
(b) Carrier wave purity	3.5.6.4 and 4.4.2.4	*See Note 1

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Table III - Production Control Inspection (Continued)

Inspection	Inspection paragraph	AQL's
5. Receiver		
(a) Frequency	3.5.7.3	6.5
(b) Undesired radiated power	3.5.7.13	*See Note 1

*Note 1: These tests shall be performed on one sample of each unit from the first month's production and on one sample of each unit from each successive 50 units produced or fraction thereof.

4.4.2.1 Power supply.- Tests shall be performed on the power supply that will verify correct operation of the control unit amplifiers at 105 and 125 volts, 60 Hz alternating current input.

4.4.2.2 Power supply.- Tests shall be performed on the power supply that will verify correct operation of all auxiliary control circuitry at 105 and 125 volts, 60 Hz alternating current input.

4.4.2.3 Carrier wave purity.- Tests shall be performed for spurious radiation. Spurious radiations shall be attenuated at least 30 dB below the carrier. The fundamental output shall be free from switching noise.

4.4.2.4 Carrier wave purity.- Tests shall be performed for spurious radiation. Spurious radiations shall be attenuated at least 20 dB below the carrier. The fundamental output shall be free from switching noise.

4.4.2.5 Rejected lots.- If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.6 Reinspection of conforming production control test units.- Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and passed production inspection after repair of all damage.

4.4.2.7 Standard test conditions.- Unless otherwise specified, standard test conditions as specified in 3.4.1.1; 3.5.1.1; and 3.6.1.1 shall be used in performing tests for each appropriate unit.

4.4.3 Environmental tests.- Environmental tests shall consist of the tests approved in accordance with 4.1.3 and shall include the tests specified in table IV.

TABLE IV - Environmental Tests

Requirement	Reference paragraph
Radio set, fixed shipboard	
1. Temperature	3.4.1.2 & 3.4.5.15
2. Shock	3.4.1.3
3. Vibration	3.4.1.4
4. Accelerated life test	See MIL-E-16400
5. Enclosure	3.4.1.6 and 3.4.4

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Table IV - Environmental Tests (continued)

Requirement	Reference paragraph
Radio set helmet	
1. Temperature	3.5.1.2 & 3.5.6.12
2. Shock	See MIL-E-16400
3. Vibration	See MIL-E-16400
4. Accelerated life test	See MIL-E-16400
5. Enclosure	3.5.1.6
Radio set handset	
1. Temperature	3.6.1.2 & 3.5.6.12
2. Shock	3.6.1.3
3. Vibration	3.6.1.3
4. Accelerated life test	See MIL-E-16400
5. Enclosure	3.6.1.4

4.4.3.1 Environmental sample units.- One Radio set, Fixed Shipboard; one Radio Set Helmet; and one Radio Set Handset shall be selected from the first month's production and subjected to the environmental tests. One sample of each shall be selected from each successive 100 units produced or fraction thereof.

4.4.3.2 If a sample unit fails the test specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and the like. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officers (see 6.4).

4.4.3.3 Reinspection of conforming environmental test units.- Unless otherwise specified, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Reliability testing.- The reliability of each end item (see 3.2.6.1) that comprises the AN/SRC-22(V) Flight deck communication system shall be verified and reported in accordance with MIL-STD-781,

4.5.1 Demonstration phase of production reliability tests.- The demonstration phase of reliability tests shall be conducted with test plan IV at the test level A except as specified below:

(a) Temperature: 25° + 10° -5°C
(b) Humidity: ambient room
(c) Vibration: any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of + 3G's.

The direction of vibration shall be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. The sample size shall be two Radio Sets, Fixed Shipboard; ten Radio Set Helmets; and ten Radio Set Handsets, selected from the first month's production. The reliability requirements of the radio set, fixed shipboard, radio set, helmet, and radio set, handset shall be individually demonstrated prior to the shipment of any production equipment.

4.5.2 Production acceptance (sampling) phase of production reliability tests.- The production acceptance (sampling) phase of reliability tests shall be conducted with test plan IV-A at test level A-1, MIL-STD-781. The sample size shall be as specified above from each subsequent month's production. The reliability tests for the radio set, shipboard, radio set, helmet, and radio set, handset shall be individually successfully completed prior to the shipment of any units of the lot from which the sample was taken.

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4.5.3 Reliability test plan.- The contractor(s) shall prepare and submit detailed reliability test plan(s) in accordance with MIL-STD-785 to the command or agency concerned (see 6.1.2). Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the command or agency concerned. No preventative maintenance shall be performed during a reliability test. Approval of the plan(s) by the command or agency concerned must be received by the contractor(s) prior to the start of reliability testing.

4.5.4 Failure notification.- The command or agency concerned shall be notified by telephone or TWX within 24 hours of failures occuring in test specified in 4.5.1 and 4.5.2 requiring repair or replacement of functional parts or assemblies.

4.5.5 Failure.- Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Only relevant failures shall be counted during the reliability test for an accept or reject decision.

4.5.6 Final restoration and disposition of tested equipment.- Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.5.7 Reliability test reports.- The contractor(s) shall submit reports that define the results of the test at the completion of each phase of reliability testing for approval by the command or agency concerned (see 6.1.2).

4.6 Inspection of preparation for delivery.- Inspection of preparation for delivery shall be in accordance with section 5. First packs which contain level A packages shall be subjected to rough handling tests (see 5.1.1).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning.- Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application.- None.

5.1.1.4 Unit packaging.- Each flight deck communications system shall be individually packaged in accordance with MIL-P-116, insuring compliance with the general paragraph under Methods of preservation (Unit protection) and the Physical protection paragraph therein and meet the rough handling tests of MIL-P-116. Cushioning and dunnage shall be used to prevent physical or mechanical damage.

5.1.1.4.1 Components.- The speaker, control unit, and base station unit of the fixed shipboard set, and the transmitters and receivers of the helmet set shall be packaged in accordance with Method II of MIL-P-116. Desiccant shall conform to MIL-D-3464, Type II or III. The remaining components and hardware shall be packaged in accordance with Method III of MIL-P-116. Items not securely affixed in place,

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such as electron tubes, shall be removed and packaged individually in transparent packages in accordance with MIL-E-17555.

5.1.1.4.2 Equipment repair parts.- Equipment repair parts or running spares shall be individually packaged in sealed transparent packages in accordance with MIL-E-17555. These items shall be packaged separately from other items within the pack in containers conforming to PPP-B-636 and identified to the applicable equipment. Repair parts lists shall be provided in accordance with MIL-E-17555.

5.1.1.4.3 Technical literature.- Technical literature shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4 mil thick, and shall not be placed within the barrier used to package any other items.

5.1.2 Level C.- Each complete flight deck communications system shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.1). The set shall be packed in the prone position.

5.2.1 Level A.- Flight deck communications systems, packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, Type overseas, style optional. Packing containers shall have closed and sealed case liners conforming to MIL-L-10547, and shall be strapped in accordance with the appendix to PPP-B-601. Containers exceeding 250 pounds gross weight shall be modified to include a skid-type base.

5.2.2 Level B.- Flight deck communications systems, packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, or PPP-B-576, Class 2.

5.2.3 Level C.- Items, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Unitized loads.- When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking.- In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1). Marking shall include set or assembly markings, and special marking for Method II packs.

5.5 General.- Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Ordering data: Procurement documents should specify the following:

6.1.1 Procurement requirements.-

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Levels of preservation-packaging and packing, and applicable marking (see section 5).

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6.1.2 Contract data requirements.- Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Reliability program plan (see 3.2.6.2)
(b) Reliability prediction (see 3.2.6.4)
(c) Reliability reports (see 3.2.6.6)
(d) Test procedures (see 4.1.3)
(e) First article inspection report (see 4.3.1)
(f) Reliability testing plan (see 4.5.3)
(g) Reliability test reports (see 4.6.7)
(h) Test points (see 3.3.3)

6.2 First article.-

6.2.1 Invitations for bids should provide that the command or agency concerned reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Definitions.-

6.3.1 Sensitivity.- The sensitivity of a receiver is the minimum value of the standard test input signal from a standard input signal source which when modulated with ±5 kHz deviation will produce at least 50 percent of the receivers rated audio power output with 12 dB signal + noise to noise ratio.

6.3.2 Modulation acceptance bandwidth.- The modulation acceptance bandwidth of the receiver is a measure of the deviation that the receiver will accept at an rf signal level 6 dB above the rf signal level limit for sensitivity.

6.3.3 Audio squelch sensitivity.- The audio squelch sensitivity of the receiver is the minimum value of the standard test input signal source, which, when modulated at 1000 Hz with a deviation of ±5 kHz will open the receiver squelch.

6.3.4 Audio power output.- Audio power output denotes the power that the receiver will deliver to a standard load under standard test conditions with ±7.5 kHz deviation.

6.3.5 Audio frequency response.- Audio frequency response denotes the degree of closeness to which the audio output of a receiver over a given continuous frequency range matches the output at 1000 Hz.

6.3.6 Hum and noise ratio.- Hum and noise ratio denotes the ratio of residual receiver audio output to rated audio output.

6.4 Environmental testing.- Approval to ship may be withheld at the discretion of the command or agency concerned, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.5 Interchangeability.- All equipments, assemblies and modules shall be electrically, mechanically and functionally interchangeable with the corresponding equipments, assemblies and modules of the Government furnished AN/SRC-22(V).

6.6 Changes from previous issue.- The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes form the previous issue of this document.

Preparing activity:
Navy - EC

(Project 5820-N526)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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